

BRK (00)		Cycles: 7	Size: 2 *
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	** Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand.	
2.1		Inc. PC .	
2.2	*** Write to u8(S.L +0) \$0100	Push PC.H onto stack.	
3.1			
3.2	*** Write to u8(S.L -1) \$0100	Push PC.L onto stack.	
4.1		Use Interrupt flags to decide which vector to use. RESET = \$FFFC. NMI = \$FFFA. IRQ/BRK = \$FFFE.	
4.2	*** Write to u8(S.L -2) \$0100	Push P onto stack with B flag if software BRK.	
5.1		Dec. S by 3.	
5.2	Read Vector	Store to Address.L.	
6.1		Set I , unset D .	
6.2	Read Vector + 1	Store to Address.H.	
7.1		Enable writes to PC (disabled by NMI/IRQ). Copy Address to PC. Set PB to 0.	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

* Concerning the BRK instruction, you should also note that although its second byte is basically a “don’t care” byte – that is, it can have any value - the BRK (and COP instruction as well) is a two-byte instruction, the second byte sometimes is used as a signature byte to determine the nature of the BRK being executed. When an RTI instruction is executed, control always returns to the second byte past the BRK opcode. — assembly-programming-manual-for-w65c816.pdf

** If NMI, IRQ, or RESET, OpCode is forced to 0 and writes to PC are disabled.

*** If handling a RESET, the writes turn into reads. The databus is populated but ignored.

ORA (01) Cycles: 6		Size: 2
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1	Read Address	Store as Pointer.
4.2		
5.1	Read Address + 1	Store as Pointer.H.
5.2		
6.1	Read Pointer:DB	Store as Operand.
6.2		
7.1		
7.2	Read PC:PB	Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.
		Store as OpCode.
+X.1		

COP (02)	Cycles: 7	Size: 2
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Write to u8(S .L+0) \$0100	Push PC .H onto stack.
3.1		
3.2	Write to u8(S .L-1) \$0100	Push PC .L onto stack.
4.1		Set Vector to \$FFF4.
4.2	Write to u8(S .L-2) \$0100	Push P onto stack.
5.1		Dec. S by 3.
5.2	Read Vector	Store to Address.L.
6.1		Set I , unset D .
6.2	Read Vector + 1	Store to Address.H.
7.1		Copy Address to PC . Set PB to 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (03)		Cycles: 4	Size: 2
Stack Relative (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Address to Pointer + (S .L \$100).	
2.2			
3.1			
3.2	Read Address	Store as Operand.	
4.1	Read PC:PB	Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.	
4.2		Store as OpCode.	
+X.1			

TSB (04)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set Z based off (A.L & Operand.L). Perform Operand = A .
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (05)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

ASL (06)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set C based off (Operand.L & \$80). Perform Operand.L <=< 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (07)		Cycles: 6	Size: 2
Direct Indirect Long (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1			
5.2	Read Address + 2	Store as Bank.	
6.1			
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

PHP (08)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S .L \$0100)+0)	Inc. PC .
1.2		
2.1		Sets Operand.L to P with X and M flags set.
2.2		Push Operand.L onto stack.
3.1		Dec. S by 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (09)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Operand.	
2.1		Inc. PC . Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .	
2.2		Store as OpCode.	
+X.1			

ASL (0A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set C based off (A.L & \$80). Perform A.L <= 1. Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PHD (0B)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to u8(S .L+0) \$0100 Write to ((S .L \$0100)-1) Read PC:PB	Inc. PC .
1.2		
2.1		
2.2		Push D .H onto stack.
3.1		
3.2		Push D .L onto stack.
4.1		
4.2		Store as OpCode.
+X.1		

TSB (0C)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set Z based off (A .L & Operand.L). Perform Operand = A .
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (0D)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

ASL (0E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set C based off (Operand.L & \$80). Perform Operand.L <=< 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (0F)		Cycles: 5	Size: 4
Absolute Long (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Address.L.	
3.1		Inc. PC .	
3.2	Read PC:PB	Stores as Bank.	
4.1		Inc. PC .	
4.2	Read Address:Bank	Store as Operand.	
5.1		Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

BPL (10)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if N is 0.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

ORA (11)		Cycles: 5	Size: 2
Direct Indirect Indexed (d),y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).	
5.2			
6.1			
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

ORA (12)		Cycles: 5	Size: 2
Direct Indirect (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1			
5.2	Read Pointer: DB	Store as Operand.	
6.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .	
6.2	Read PC:PB	Store as OpCode.	
+X.1			

ORA (13)	Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$100.)
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.
7.2	Read PC:PB .	Store as OpCode.
+X.1		

TRB (14)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set Z based off (A.L & Operand.L). Perform Operand &= ~ A .
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (15)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB	Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2		Store as OpCode.
+X.1		

ASL (16)	Cycles: 6	Size: 2
Direct, X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Set C based off (Operand.L & \$80). Perform Operand.L <= 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.1		
6.2	Write to Address	Write Operand.L.
7.1	Read PC:PB	
7.2		Store as OpCode.
+X.1		

ORA (17) Cycles: 6		Size: 2
Direct Indirect Indexed Long [d],y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

CLC (18)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Sets the C flag to 0.
2.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (19)	Cycles: 4	Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB .	Store as OpCode.
+X.1		

INC (1A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform A.L += 1, set N based off (A.L & \$80), set Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

TSC (1B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Discard.
2.1		Sets S to A .
2.2		Store as OpCode.
+X.1		

TRB (1C)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set Z based off (A .L & Operand.L). Perform Operand &= ~ A .
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (1D) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB .	Store as OpCode.
+X.1		

ASL (1E)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set C based off (Operand.L & \$80). Perform $\text{Operand.L} \ll= 1$. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB .	Store as OpCode.
+X.1		

ORA (1F) Cycles: 5		Size: 4
Absolute Long, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read PC:PB .	Stores as Bank.
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB .	Store as OpCode.
+X.1		

JSR (20)	Cycles: 6	Size: 3
Absolute		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read PC:PB	Discard.
4.1		Dec. PC .
4.2	Write to u8(S .L+0) \$0100	Push PC .H onto stack.
5.1		
5.2	Write to u8(S .L-1) \$0100	Push PC .L onto stack.
6.1		Dec. S by 2. Perform PC = Address.
6.2	Read PC:PB	Store as OpCode.
+X.1		

AND (21)		Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB .	Store as Pointer.	
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.	
2.2		If D .L is 0, skip the next cycle.	
3.1		Address.H = Pointer.H (fixes high byte of address).	
3.2			
4.1			
4.2	Read Address	Store as Pointer.	
5.1			
5.2	Read Address + 1	Store as Pointer.H.	
6.1			
6.2	Read Pointer: DB	Store as Operand.	
7.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.	
7.2	Read PC:PB .	Store as OpCode.	
+X.1			

JSL (22)	Cycles: 8	Size: 4
Absolute Long		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Write to u8(S .L+0) \$0100	Push PB onto stack.
4.1		
4.2	Read u8(S .L+0) \$0100	Discard.
5.1		
5.2	Read PC:PB	Stores as Bank.
6.1		
6.2	Write to ((S .L \$0100)-1)	Push PC .H onto stack.
7.1		
7.2	Write to ((S .L \$0100)-2)	Push PC .L onto stack.
8.1		Dec. S by 3. Copy Address to PC . Copy Bank to PB .
8.2	Read PC:PB	Store as OpCode.
+X.1		

AND (23)	Cycles: 4	Size: 2
Stack Relative (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB .	Store as OpCode.
+X.1		

BIT (24)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set Z based off (A.L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
4.2	Read PC:PB	Store as OpCode.
+X.1		

AND (25)		Cycles: 3	Size: 2
Direct (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Operand.	
4.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
4.2	Read PC:PB	Store as OpCode.	
+X.1			

ROL (26)		Cycles: 5	Size: 2
Direct (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.L.	
5.1		Set C based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1) C . Set N based off (Operand.L & \$80) and Z based off Operand.L.	
5.2	Write to Address	Write Operand.L.	
6.1			
6.2	Read PC:PB	Store as OpCode.	
+X.1			

AND (27)	Cycles: 6	Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

PLP (28)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S by 1.
3.2		Store as Operand.L.
4.1	Read S.L \$0100	P = Operand.L, set X and M flags, set X.H and Y.H to 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

AND (29)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

ROL (2A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set C based off (A.L & \$80). Perform A.L = (A.L << 1) C . Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PLD (2B)	Cycles: 5	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read ((S .L \$0100)+1)	Inc. PC .
1.2		
2.1		
2.2		
3.1		
3.2		Store as Operand.L.
4.1		
4.2		Store as Operand.H.
5.1		Inc. S by 2. Set D to Operand. Set N based off (D .H & \$80). Set Z based off D .
5.2		Store as OpCode.
+X.1		

BIT (2C)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set Z based off (A .L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
4.2	Read PC:PB	Store as OpCode.
+X.1		

AND (2D)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

ROL (2E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set C based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1) C . Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

AND (2F)		Cycles: 5	Size: 4
Absolute Long (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Address.L.	
3.1		Inc. PC .	
3.2	Read PC:PB	Stores as Bank.	
4.1		Inc. PC .	
4.2	Read Address:Bank	Store as Operand.	
5.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

BMI (30)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if N is 1.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

AND (31) Cycles: 5		Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

AND (32)	Cycles: 5	Size: 2
Direct Indirect (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer: DB	Store as Operand.
6.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
6.2	Read PC:PB	Store as OpCode.
+X.1		

AND (33)		Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB .	Store as Pointer.	
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).	
2.2			
3.1			
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1		Perform Tmp = ui32(Pointer + Y).	
		Pointer = ui16(Tmp).	
		Bank = DB + (Tmp >> 16).	
5.2			
6.1			
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.	
7.2	Read PC:PB .	Store as OpCode.	
+X.1			

BIT (34)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB .	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB .	Set Z based off (A.L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
5.2		Store as OpCode.
+X.1		

AND (35)		Cycles: 4	Size: 2
Direct, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Operand.	
2.1		Inc. PC .	
2.2			
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.	
3.2			
4.1			
4.2	Read Address	Store as Operand.	
5.1	Read PC:PB	Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
5.2		Store as OpCode.	
+X.1			

ROL (36)	Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Set C based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1) C . Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.1		
6.2	Write to Address	Write Operand.L.
7.1	Read PC:PB	
7.2		Store as OpCode.
+X.1		

AND (37)		Cycles: 6	Size: 2
Direct Indirect Indexed Long [d],y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.	
5.1			
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.	
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).	
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

SEC (38)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Sets the C flag to 1.
2.2	Read PC:PB	Store as OpCode.
+X.1		

AND (39)		Cycles: 4	Size: 3
Absolute, Y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Pointer.H.	
3.1		Inc. PC . Perform Orig = Pointer.	
		Tmp = ui32(Pointer + Y).	
		Pointer = ui16(Tmp).	
		If Orig.H == Pointer.H, skip 2 half-cycles.	
		Bank = DB + (Tmp >> 16).	
3.2			
4.1			
4.2	Read Pointer:Bank	Store as Operand.	
5.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

DEC (3A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform A.L -= 1, set N based off (A.L & \$80), set Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

TSC (3B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set A to (S .L \$0100). Set N based off (A .H & \$80) and Z based off A .
2.2	Read PC:PB	Store as OpCode.
+X.1		

BIT (3C)	Cycles: 4	Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set Z based off (A .L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
5.2	Read PC:PB	Store as OpCode.
+X.1		

AND (3D) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.
5.2	Read PC:PB	Store as OpCode.
+X.1		

ROL (3E)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set C based off (Operand.L & \$80). Perform $\text{Operand.L} = (\text{Operand.L} \ll 1) \mid \text{C}$. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB .	Store as OpCode.
+X.1		

AND (3F)		Cycles: 5	Size: 4
Absolute Long, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB .	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC:PB .	Store as Pointer.H.	
3.1		Inc. PC .	
3.2	Read PC:PB .	Stores as Bank.	
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$.	
		Pointer = $\text{ui16}(\text{Tmp})$.	
		Bank += (Tmp >> 16).	
4.2	Read Pointer:Bank	Store as Operand.	
5.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
5.2	Read PC:PB .	Store as OpCode.	
+X.1			

RTI (40)	Cycles: 6	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S by 1.
3.2	Read ((S .L \$0100)+0)	Store as Operand.L.
4.1		Set P to ((Operand.L & ~\$30) (P & \$30)).
4.2	Read ((S .L \$0100)+1)	Store as Operand.L.
5.1		
5.2	Read ((S .L \$0100)+2)	Store as Operand.H.
6.1		Inc. S by 2. Set PC to Operand.
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (41)	Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	Read Address + 1	Store as Pointer.H.
6.1		
6.2	Read Pointer: DB	Store as Operand.
7.1		Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.
7.2	Read PC:PB	Store as OpCode.
+X.1		

WDM (42) Cycles: 2		Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (43)	Cycles: 4	Size: 2
Stack Relative (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

MVP (44)	Cycles: 7	Size: 3
Block Move (Positive)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set DB to Operand.L.
2.2	Read PC:PB	Stores as Bank.
3.1		Inc. PC .
3.2	Read X:Bank	Store as Operand.L.
4.1		
4.2	Write Y:DB	Write Operand.L.
5.1		Perform A -= 1. If X flag is set, perform X.L -= 1 and Y.L -= 1, otherwise perform X -= 1 and Y -= 1. If A is not \$FFFF, perform PC -= 3.
5.2		
6.1		
6.2		
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (45)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (46)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set C based off (Operand.L & \$01). Perform Operand.L >>= 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (47)	Cycles: 6	Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

PHA (48)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S .L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		Push A .L onto stack.
3.1		Dec. S by 1.
3.2		Store as OpCode.
+X.1		

EOR (49)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Perform A ^= Operand . Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (4A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set C based off (A.L & \$01). Perform A.L >>= 1. Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PHK (4B)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S .L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		Push PB onto stack.
3.1		Dec. S by 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

JMP (4C)	Cycles: 3	Size: 3
Absolute		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Set PC to Address.
3.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (4D)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (4E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set C based off (Operand.L & \$01). Perform Operand.L >>= 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (4F)	Cycles: 5	Size: 4
Absolute Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.L.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC .
4.2	Read Address:Bank	Store as Operand.
5.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

BVC (50)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if V is 0.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

EOR (51)	Cycles: 5	Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (52)	Cycles: 5	Size: 2
Direct Indirect (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer: DB	Store as Operand.
6.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (53)	Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.
7.2	Read PC:PB	Store as OpCode.
+X.1		

MVN (54)	Cycles: 7	Size: 3
Block Move (Negative)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set DB to Operand.L.
2.2	Read PC:PB	Stores as Bank.
3.1		Inc. PC .
3.2	Read X:Bank	Store as Operand.L.
4.1		
4.2	Write Y:DB	Write Operand.L.
5.1		Perform A -= 1. If X flag is set, perform X.L += 1 and Y.L += 1, otherwise perform X += 1 and Y += 1. If A is not \$FFFF, perform PC -= 3.
5.2		
6.1		
6.2		
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (55)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB	Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2		Store as OpCode.
+X.1		

LSR (56)	Cycles: 6	Size: 2
Direct, X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Set C based off (Operand.L & \$01). Perform Operand.L >>= 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.1		
6.2	Write to Address	Write Operand.L.
7.1	Read PC:PB	
7.2		Store as OpCode.
+X.1		

EOR (57)		Cycles: 6	Size: 2
Direct Indirect Indexed Long [d],y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.	
5.1			
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.	
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).	
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

CLI (58)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Sets the I flag to 0.
2.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (59)	Cycles: 4	Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

PHY (5A)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S .L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		Push Y .L onto stack.
3.1		Dec. S by 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

TCD (5B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set D to A . Set N based off (D .H & \$80) and Z based off D .
2.2	Read PC:PB	Store as OpCode.
+X.1		

JML (5C)	Cycles: 4	Size: 4
Absolute Long		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Copy Address to PC . Copy Bank to PB .
4.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (5D) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (5E)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set C based off (Operand.L & \$01). Perform $\text{Operand.L} \gg= 1$. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (5F)		Cycles: 5	Size: 4
Absolute Long, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Pointer.H.	
3.1		Inc. PC .	
3.2	Read PC:PB	Stores as Bank.	
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$.	
		Pointer = $\text{ui16}(\text{Tmp})$.	
		Bank += (Tmp >> 16).	
4.2	Read Pointer:Bank	Store as Operand.	
5.1		Perform $\text{A} \wedge = \text{Operand}$. Set N based off (A.L & \$80) and Z based off A.L .	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

RTS (60)	Cycles: 6	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read ((S.L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S by 1.
3.2		Store as Operand.L.
4.1		
4.2		Store as Operand.H.
5.1		
5.2		
6.1	Read PC:PB	Inc. S by 1. Set PC to (Operand + 1).
6.2		Store as OpCode.
+X.1		

ADC (61) Cycles: 6		Size: 2
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1	Read Address	Store as Pointer.
4.2		
5.1	Read Address + 1	Store as Pointer.H.
5.2		
6.1	Read Pointer: DB	Store as Operand.
6.2		
7.1		If D flag == 0: Result = ui16(A .L) + ui16(Operand) + C . V flag = (~ui16(A .L) ^ ui16(Operand)) & (ui16(A .L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A .L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A .L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A .L >> 4) ^ (Operand >> 4)) & ((A .L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2		Store as OpCode.
+X.1		

PER (62)	Cycles: 6	Size: 3
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Operand.H.
3.1		Inc. PC . Set Operand to Operand + PC .
3.2		
4.1		
4.2	Write to ((S .L \$0100)+0)	Push Operand.H onto stack.
5.1		
5.2	Write to ((S .L \$0100)-1)	Push Operand.L onto stack.
6.1		Dec. S by 2.
6.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (63)		Cycles: 4	Size: 2
Stack Relative (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).	
2.2			
3.1			
3.2	Read Address	Store as Operand.	
4.1		If D flag == 0: Result = ui16(A .L) + ui16(Operand) + C . V flag = (~ui16(A .L) ^ ui16(Operand)) & (ui16(A .L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A .L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A .L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A .L >> 4) ^ (Operand >> 4)) & ((A .L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.	
4.2		Store as OpCode.	
+X.1			

STZ (64)	Cycles: 3	Size: 2
Direct (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Write to Address	Write 0.
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		